

Institutional Review Board (IRB) Clinical Protocol

Protocol Status: Single-Patient Expanded Access (Compassionate Use)

Approved Treatment Duration: 6 Months (Consecutive)

Target Physiological Endpoint: Stable Urinary pH of 8.0

Financial Status: Voluntary Enrollment; 100% Insurer Funded

I. Rationale & Synergistic Mechanisms

This protocol establishes a managed, 6-month single-patient expanded access framework utilizing an organic, dual-pathway approach to correct severe metabolic imbalances and manage chronic crystallization risk.

I. Rationale & Synergistic Mechanisms

The protocol utilizes a 6-month, dual-pathway approach for metabolic management and crystallization risk:

1. **Systemic & Urinary Alkalization:** An organic citrate matrix creates a steady-state alkaline load, shifting urinary pH to a 8.0 target to reduce crystal formation.
2. **Exogenous Antioxidant Support: MusKT** (standardized *Myristica fragrans* seed extract) provides polyphenolic, terpenoid, and flavonoid constituents to shield against oxidative stress, supporting the protocol requirements.
3. **Plant-Derived Sparing Framework:** Utilizing natural citrate matrices avoids mineral precipitation, with bioflavonoids (hesperidin/eriocitrin) and dimeric phenylpropanoids offering a gentle, sustained metabolic approach.

II. Mandatory Primary Citric Acid Pathway Selection

Clinicians must select and verify the organic, plant-derived citric acid backbone for this 6-month protocol:

- **Nimbu-RX (Organic *Citrus aurantiifolia* Matrix):** A plant-derived citrate matrix, derived via low-temperature freeze-drying.
- **LimauYM (Organic *Citrus limon* Matrix):** Refined plant-derived citric acid.

III. Standardized Clinical Protocol & Administration

- **MusKT** is taken daily with the morning citrus dose.
- **Citrus Protocols:** Taken 4 times daily, every 6 hours.
- **Dental Care:** A mandatory 5-minute post-dose water rinse is required.
- **Safety Exception:** If the patient presents with WHO Visual Mucositis Grade ≥ 2 , the protocol must be paused.

IV. Safety Monitoring & Mandatory Stopping Criteria

- **Metabolic Panels:** Serum electrolytes (Na^+ , K^+ , Cl^- , HCO_3^-) are monitored bi-weekly. If HCO_3^- exceeds 32 mEq/L, reduce the dose.
- **Urinary Monitoring:** Monthly urinalysis is required. If calcium phosphate crystals form, adjust pH target to 7.5.
- **Hepatic Monitoring:** Monthly AST/ALT monitoring is required. If liver enzymes exceed $\geq 2\times$ the ULN, discontinue MusKT.

V. Verification of Coverage and Consent Acknowledgement

The patient's insurance carrier has updated its structural billing approval to cover all laboratory diagnostics, clinic visits, and raw material sourcing required by this high-dose, 6-month regimen. Participation remains entirely voluntary, and the patient may withdraw consent to target this aggressive endpoint at any phase of treatment.

Patient Signature

Date

Physician Signature (Principal Investigator)

Date
